



SWASTIK SUMAN NAYAK

Aspiring Data Analyst | BI & ML Enthusiast

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EDUCATION

B.Tech, Computer Science Engineering
KIIT University
2023 – 2027 | CGPA: 8.07

Senior Secondary (10+2)
Kendriya Vidyalaya, ARC CBT
2022 – 2023 | 78.9%

TECHNICAL SKILLS

- Python, Java, C
- HTML, CSS, JavaScript
- DAX (Power BI)
- SQL Fundamentals

TOOLS & PLATFORMS

- Power BI
- Git & GitHub
- VS Code
- MATLAB

CORE CONCEPTS

- Data Visualization
- Data Cleaning & Prep
- Machine Learning
- Data Structures & Algorithms

OPERATING SYSTEMS

- Windows
- Linux

INTERESTS

- Competitive Coding
- Business Intelligence & Data Viz

PROFESSIONAL SUMMARY

Computer Science Engineering student at KIIT with hands-on experience building Power BI dashboards and machine learning models. Skilled at transforming raw datasets into clear, decision-ready visual insights through DAX measures, data cleaning, and dashboard design, while also applying Python and ML fundamentals to solve real-world prediction and classification problems. Seeking opportunities in data analytics and business intelligence where strong analytical thinking and visualization skills can drive measurable business impact.

BUSINESS INTELLIGENCE PROJECTS

Employee Attendance Tracker Dashboard — Power BI

- Designed an interactive attendance-tracking dashboard using a custom CSV dataset, with a rose pink/lavender visual theme for clear stakeholder readability.
- Built calculated measures and visuals to track attendance trends, absenteeism patterns, and employee-level summaries.
- Documented the project in a full written report covering data sourcing, transformation steps, and dashboard design choices.

Superstore Sales Dashboard — Power BI

- Built a sales-performance dashboard on the Superstore dataset using a dark navy theme, covering revenue, profit, and category-level breakdowns.
- Applied data cleaning and transformation in Power Query to prepare raw sales data for accurate visualization.
- Designed interactive filters and visuals to let stakeholders drill down by region, category, and time period.

HR Attrition & Retention Analysis Dashboard — Power BI

- Developed a two-page dashboard analyzing employee attrition and retention using the IBM HR Analytics dataset.
- Authored 8 custom DAX measures to calculate attrition rate, average tenure, and department-level retention metrics.
- Debugged and refined visuals through iterative testing, and produced a complete written project report documenting methodology and findings.

MACHINE LEARNING PROJECTS

AEGIS — Autonomous Digital Immune System | Hackathon Project

Tech stack: Python, FastAPI, LangGraph, React, WebSockets, ChromaDB, NetworkX

- Engineered a 6 agent AI security pipeline using LangGraph to autonomously detect, analyze and respond to cyber threats in real-time.
- Built a live WebSocket streaming architecture between FastAPI backend and React frontend, enabling token-by-token agent reasoning and dynamic graph updates.
- Integrated ChromaDB vector memory for incident recall, allowing the system to recognize recurring attack patterns and auto-suggest containment strategies.
- Developed an interactive 28-node digital twin using NetworkX graph traversal, dynamically highlighting compromised assets, lateral movement paths, and blast radius.
- Implemented MITRE ATT&CK mapping, risk scoring algorithms, and Web Speech API voice commands for hands-free SOC operations.

Sentiment Analysis Using Speech and Facial Recognition

Tech stack: Python, Audio/Facial Feature Extraction, Machine Learning

- Developed a multimodal system to detect human emotion by combining speech-signal analysis with facial expression recognition.
- Engineered audio and visual feature-extraction pipelines to classify sentiments such as happy, sad, angry, and neutral.
- Fused speech and visual signals to improve real-time emotion-recognition accuracy over single-modality approaches.

AI-Based Fact Checker Using Inverse Scaling Laws

Tech stack: Python, NLP, Machine Learning

- Designed an AI-driven fact-checking pipeline to identify misinformation and false claims in text.
- Applied inverse scaling law concepts to analyze model behavior when detecting complex, misleading statements.
- Built a text-analysis and credibility-scoring pipeline to support claim verification.

HACKATHONS & COMPETITIONS

Decodex Hackathon — organized under NLD Synapse

- Collaborated in a team to design and prototype an AI-based solution within a limited time frame.
- Contributed to problem-solving, rapid prototyping, and model development under hackathon time constraints.
- Gained experience presenting a technical solution to judges and peers.

HackWithInfy — Round 2 (Coding Round)

- Solved algorithmic problems involving greedy approaches, heap-based structures, and Kadane's algorithm variants under timed conditions.
- Built a stronger working framework for choosing the right data structure based on problem constraints.

OTHER TECHNICAL ACTIVITIES

- Built a FRIDAY AI voice assistant in Python, integrating GPT-4o for conversational responses on a Windows desktop environment.
- Developed two Doctor Strange–inspired AR hand-tracking web applications using MediaPipe Hands and HTML/Canvas, with gesture-triggered visual effects.
- Completed coursework projects including inventory management systems and data analysis tools using Java and Python.
- Applied MATLAB for signal and image processing in academic and self-driven projects.